Speech Understanding
Assignment 2

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Abstract

This paper presents a comprehensive modular speech processing pipeline for Hinglish (mixed Hindi-English) lecture audio addressing three critical challenges in multilingual speech technology. We formulate the N-gram language model scoring mechanism with Laplace smoothing and demonstrate its integration into Whisper's beam search decoding. We conduct a detailed ablation study comparing Dynamic Time Warping (DTW) prosody transfer against linear monotonic warping, demonstrating a 2.1 dB reduction in Mel-Cepstral Distortion. Our frame-level Language Identification model achieves 96.87% accuracy with 98.36% F1 on Hindi frames, employing a CNN + BiLSTM architecture. We provide mathematical formulations for memory-safe DTW with fallback strategies for long-form audio (¿10 minutes). Experimental validation on 10-minute lecture segments demonstrates end-to-end pipeline robustness across 42 translation chunks with consistent prosody synthesis quality.

Code Availability: The implementation of the proposed pipeline is publicly available at:
https://github.com/viveksapkal2793/Speech-Pipeline

1 Introduction

Hinglish, the colloquial mixing of Hindi and English prevalent in Indian academic and professional settings, poses unique challenges for automatic speech recognition (ASR) and text-to-speech (TTS) synthesis systems. Traditional ASR systems optimized for monolingual data struggle with code-switching phenomena—the seamless alternation between languages at the sentence, phrase, or word level [1]. Similarly, synthesized speech often lacks naturalness when prosody (pitch, energy, duration) from the source speaker is not preserved.

This work addresses three interconnected problems:

1. **ASR Accuracy Under Language Mixing**: N-gram language models can bias decoding toward domain-specific phrases and maintain language generality through Laplace smoothing and backoff strategies [2].
2. **Naturalness via Prosody Transfer**: Synthesized speech requires prosody transfer from reference audio to avoid the "TTS monotone" effect. We employ Dynamic Time Warping (DTW) for temporal alignment with memory-safe fallback for long-form audio.
3. **Code-Switching Detection**: Frame-level language identification enables the detection of language-switch boundaries at the acoustic level, necessary for linguistic analysis and downstream processing [3].

The pipeline integrates state-of-the-art components: Whisper Small [4] for ASR, IndicTrans2 [5] for translation, Parler TTS for synthesis, and a custom CNN+BiLSTM architecture for Language Identification. Each module is independently trained and validated, then orchestrated into an end-to-end system capable of processing 10-minute lecture segments with consistent quality.

2 Related Work

2.1 Language Identification and Code-Switching

Frame-level language identification has been extensively studied [6, 7]. Traditional approaches use Gaussian Mixture Models (GMMs) over MFCC features; more recent work employs deep learning architectures. For code-switching at word level or utterance level, fusion of language models and acoustic classifiers has proven effective [8]. Our frame-level CNN+BiLSTM approach builds on this foundation, targeting temporal precision for boundary detection.

2.2 N-gram Language Models and Beam Search

Kneser-Ney smoothing [2] is a widely-used technique for n-gram language models, improving upon simple Laplace smoothing through continuation probabilities. In ASR decoding, language models are typically integrated via weighted rescoring of acoustic hypotheses [9]. Modern

ASR systems like Whisper [4] employ multilingual pre-training but benefit from domain-specific LM biasing in constrained scenarios.

2.3 Prosody Transfer and Synthesis

Prosody transfer in TTS has been addressed through multiple paradigms: global speaker adaptation [10], fine-grained prosody prediction from text [12], and reference-based transfer [11]. Dynamic Time Warping (DTW) [**?**] is a classical signal processing technique for temporal alignment, widely used in speech applications. Our contribution focuses on memory-safe DTW with graceful linear fallback for realistic long-form audio scenarios.

3 Mathematical Formulation

3.1 N-gram Language Model Scoring

Given a sequence of words $w_1, w_2, \ldots, w_n$, the n-gram language model decomposes the joint probability as a product of conditional probabilities:

$$P(w_1, \ldots, w_n) = \prod_{i=1}^{n} P(w_i | w_{i-n+1:i-1}) \tag{1}$$

where $w_{i-n+1:i-1}$ denotes the preceding context of $n-1$ words. For a bigram model ($n = 2$), we have:

$$P(w_1, \ldots, w_n) = \prod_{i=1}^{n} P(w_i | w_{i-1}) \tag{2}$$

For a trigram model ($n = 3$):

$$P(w_1, \ldots, w_n) = \prod_{i=1}^{n} P(w_i | w_{i-2:i-1}) \tag{3}$$

3.1.1 Laplace Smoothing

To handle unseen n-grams (sparse data problem), we apply Laplace (add-one) smoothing:

$$P_{\text{smooth}}(w_i | c) = \frac{\text{count}(w_i, c) + 1}{\text{count}(c) + |V|} \tag{4}$$

where:

- $\text{count}(w_i, c)$ = number of times word w_i follows context c in the corpus
- $\text{count}(c)$ = number of times context c appears in the corpus
- $|V|$ = vocabulary size (number of unique words)

This ensures $P_{\text{smooth}}(w_i|c) > 0$ for all w_i and c, preventing zero probabilities.

3.1.2 Log-Probability Aggregation

For computational stability and to enable additive scoring in beam search, we work in log space. The composite score for a word sequence is:

$$\text{score}(w_1 : n) = \sum_{i=1}^{n} \log P_{\text{smooth}}(w_i | w_{i-k:i-1}) \tag{5}$$

where $k = \text{order} - 1$ (e.g., $k = 2$ for trigrams). This formulation allows incremental scoring: when extending a partial hypothesis with word w_i, we add $\log P_{\text{smooth}}(w_i|\text{context})$ to the accumulated score.

3.1.3 Context Padding

In our implementation, we pad sequences with special tokens $\langle s \rangle$ (start) and $\langle /s \rangle$ (end):

$$\text{padded} = [\langle s \rangle]^{n-1} + w_1 : n + [\langle /s \rangle] \tag{6}$$

This ensures all n-grams are well-defined at sequence boundaries and allows the model to learn meaningful start/end statistics.

3.2 Beam Search with Language Model Rescoring

During Whisper decoding, we maintain B active hypotheses (beams). At each decoding step, for each hypothesis, Whisper produces logits over the token vocabulary. We combine acoustic scores with language model scores using a weighted linear combination:

$$\text{score}_j = L_a^{(j)} + \alpha \cdot S_{\text{lm}}(j) - \beta \cdot |t_j| \tag{7}$$

where:

- $L_a^{(j)}$ = log-probability assigned by Whisper's acoustic model for token j
- $S_{\text{lm}}(j)$ = log-probability from the n-gram language model (Equation 5)
- α = LM weight hyperparameter (empirically set to 0.25 in our pipeline)
- β = length penalty weight (set to 0.0 to avoid bias toward shorter sequences)
- $|t_j|$ = length (number of tokens) in hypothesis j

The length penalty term $\beta \cdot |t_j|$ prevents beam search from favoring extremely short hypotheses; setting $\beta = 0.0$ indicates no length bias.

Algorithm 1 Constrained Whisper Beam Search

```
Initialize: beams ← single hypothesis with initial tokens
for each decoding step do
    for each active beam do
        Compute acoustic logits L_a from Whisper
        For each candidate token j:
            Extend hypothesis with token j
            Compute S_lm for extended sequence
            Compute combined score via Eq. 7
    end for
    Select top B hypotheses by combined score
    Remove any hypotheses that reach end-of-sequence token
end for
Return best hypothesis with highest score
```

3.2.1 Beam Search Algorithm

3.3 Prosody Transfer Module

3.3.1 Feature Extraction

We extract two complementary prosody features: Fundamental Frequency (F0) and energy.

F0 Extraction (Pitch): Using the YIN algorithm [13]:

$$F_0(t) = \arg\min_f \sum_{d=1}^{D} |y(t) - y(t+d)|^2 \quad \forall f \in [50, 500] \text{ Hz} \tag{8}$$

where $y(t)$ is the audio signal and D is the maximum lag in samples. YIN is more robust to octave errors than autocorrelation-based methods.

Energy Extraction: We compute RMS energy over fixed-length frames:

$$E(t) = \sqrt{\frac{1}{N} \sum_{i=0}^{N-1} y(t \cdot h + i)^2} \tag{9}$$

where:

- h = hop length (320 samples at 16 kHz = 20 ms frame shift)
- N = frame length (typically $N = 2 \times h = 640$ samples at 16 kHz = 40 ms)

Both $F_0(t)$ and $E(t)$ are computed on the same time grid for alignment.

3.3.2 Feature Stack Representation

To align both features jointly, we stack them into a 2-D matrix:

$$S_{\text{stack}}(t) = \begin{bmatrix} \log(1 + F_0(t)) \\ \log(1 + E(t)) \end{bmatrix} \tag{10}$$

The log transform stabilizes features with skewed distributions and makes DTW distance more symmetric.

3.3.3 Dynamic Time Warping (DTW)

DTW aligns two sequences of different lengths by finding a monotonic path through their joint cost matrix that minimizes total distance. Given source contours $X \in \mathbb{R}^{T_x \times 2}$ and target contours $Y \in \mathbb{R}^{T_y \times 2}$ (both with 2 features), the accumulated cost matrix $D \in \mathbb{R}^{T_x \times T_y}$ is computed recursively:

$$D[i,j] = d(X[i], Y[j]) + \min(D[i-1,j], D[i,j-1], D[i-1,j-1]) \tag{11}$$

where $d(\cdot,\cdot)$ is the Euclidean distance:

$$d(X[i], Y[j]) = \sqrt{(X[i,0] - Y[j,0])^2 + (X[i,1] - Y[j,1])^2} \tag{12}$$

Boundary conditions:

$$D[0,0] = d(X[0], Y[0]) \tag{13}$$

$$D[i,0] = D[i-1,0] + d(X[i], Y[0]) \quad \forall i > 0 \tag{14}$$

$$D[0,j] = D[0,j-1] + d(X[0], Y[j]) \quad \forall j > 0 \tag{15}$$

The final DTW distance is $D[T_x - 1, T_y - 1]$.

3.3.4 Warp Path Extraction

After computing D, we backtrack from $D[T_x - 1, T_y - 1]$ to $D[0,0]$ to recover the alignment path. This yields index pairs $(i_k, j_k)_{k=1}^{K}$ mapping source frames to target frames.

3.3.5 Memory Guard and Linear Fallback

DTW requires allocating a matrix of size $T_x \times T_y$. For long audio, this becomes prohibitive. We guard allocation with:

$$\text{cells} = T_x \times T_y \tag{16}$$

If cells $> M_{\max}$ (20 million in our implementation), we fall back to linear monotonic warping, which avoids matrix allocation entirely.

3.3.6 Linear Monotonic Warping (Fallback)

When DTW is infeasible, we approximate via uniform linear interpolation:

$$\text{warp_path}[j] = \left\lfloor \frac{j \cdot (T_x - 1)}{T_y - 1} \right\rfloor \quad \forall j \in [0, T_y - 1] \tag{17}$$

This monotonic path approximates optimal DTW alignment in $O(T_y)$ time vs. $O(T_x \cdot T_y)$ for full DTW, with negligible memory overhead.

3.3.7 Prosody Application

After computing the warp path, we align source and target features and apply transformations.

Pitch Shift (Semitones):

We compute the median ratio between aligned source and target F0 values (considering only voiced frames where $F_0 > 0$):

$$\text{pitch_ratio} = \frac{\text{median}(F_{0,\text{src}}[i_k])}{\text{median}(F_{0,\text{tgt}}[j_k])} \tag{18}$$

We clip this ratio to $[0.5, 2.0]$ to avoid extreme pitch shifts, then convert to semitones:

$$\text{shift}_{\text{semitones}} = 12 \cdot \log_2(\text{clip}(\text{pitch_ratio}, 0.5, 2.0)) \tag{19}$$

Energy Scaling:

We apply frame-wise energy adjustment:

$$y'(t) = y(t) \cdot e^{E_{\text{src}}(t) - E_{\text{tgt}}(t)} \tag{20}$$

where $E_{\text{src}}(t)$ and $E_{\text{tgt}}(t)$ are the log-energy contours after alignment via the warp path.

4 Language Identification and Code-Switching

4.1 Model Architecture

Our frame-level Language Identification model processes acoustic features (LFCC) through a CNN feature extractor followed by a BiLSTM temporal encoder and a classification head:

$$\text{logits}(t) = \text{Head}(\text{BiLSTM}(\text{CNN}(\text{features}(t)))) \tag{21}$$

4.1.1 CNN Feature Extractor

The CNN feature extractor reduces spectral dimensionality while preserving temporal structure. Pooling is applied only along the frequency axis (kernel size 2 in dimension 1, 1 in dimension 3) to maintain frame-level temporal resolution.

4.1.2 BiLSTM Temporal Encoder

After CNN, we process the temporal sequence bidirectionally:

$$\vec{h}_t, \overleftarrow{h}_t = \text{BiLSTM}(x_t, \vec{h}_{t-1}, \overleftarrow{h}_{t-1}) \tag{22}$$

$$h_t = [\vec{h}_t; \overleftarrow{h}_t] \in \mathbb{R}^{2 \cdot d_{\text{hidden}}} \tag{23}$$

where $d_{\text{hidden}} = 128$ (hidden size), and $[\cdot;\cdot]$ denotes concatenation.

Algorithm 2 CNN Architecture for LID

Input: features $\in \mathbb{R}^{B \times T \times F}$ (batch, time, LFCC features)
features $\leftarrow$ features.unsqueeze(1) $\in \mathbb{R}^{B \times 1 \times T \times F}$
Conv2d(1, 32, kernel_size=3, padding=1)
BatchNorm2d(32) + ReLU
Conv2d(32, 32, kernel_size=3, padding=1)
BatchNorm2d(32) + ReLU
MaxPool2d(2, 1) # Pool over frequency, preserve time
Conv2d(32, 64, kernel_size=3, padding=1)
BatchNorm2d(64) + ReLU
MaxPool2d(2, 1)
Conv2d(64, 128, kernel_size=3, padding=1)
BatchNorm2d(128) + ReLU
MaxPool2d(2, 1)
features $\leftarrow$ *AdaptiveAvgPool2d*$((1, T)) \in \mathbb{R}^{B \times 128 \times 1 \times T}$
features $\leftarrow$ features.squeeze(2).transpose(1,2) $\in \mathbb{R}^{B \times T \times 128}$
Output: features $\in \mathbb{R}^{B \times T \times 128}$

4.1.3 Classification Head

$$\text{logits}_t = \text{Linear}(\text{Dropout}(h_t)) \tag{24}$$

where *Dropout*$(p = 0.2)$ is applied before the final linear layer projecting to $C = 2$ classes (English, Hindi).

4.2 Frame-Level Metrics

For N test frames with ground-truth labels y_i and predictions $\hat{y}_i$ (both in $\{0, 1\}$ for binary English/Hindi), we compute:

4.2.1 Accuracy

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}[\hat{y}_i = y_i] \tag{25}$$

4.2.2 Per-Class Precision and Recall

For class $c \in \{0, 1\}$:

$$\text{Precision}_c = \frac{\sum_i \mathbb{1}[\hat{y}_i = c \text{ and } y_i = c]}{\sum_i \mathbb{1}[\hat{y}_i = c]} \tag{26}$$

$$\text{Recall}_c = \frac{\sum_i \mathbb{1}[\hat{y}_i = c \text{ and } y_i = c]}{\sum_i \mathbb{1}[y_i = c]} \tag{27}$$

4.2.3 F1-Score

$$\text{F1}_c = 2 \cdot \frac{\text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \tag{28}$$

4.2.4 Macro-F1

$$\text{Macro-F1} = \frac{1}{2} \sum_{c=0}^{1} \text{F1}_c \tag{29}$$

4.3 Confusion Matrix Representation

From our evaluation on the 10-minute LID eval segment (24,409 frames):

True / Pred.	English	Hindi	Recall	Total Frames
English	854	145	0.8546	999
Hindi	695	23,715	0.9713	24,410
Precision	0.5511	0.9942	—	—

Table 1: Evaluated LID Confusion Matrix

The high Hindi precision (99.42%) and recall (97.13%) indicate strong Hindi identification. However, English precision is notably lower (55.11%), suggesting the model tends to misclassify English frames as Hindi. This class imbalance likely reflects the training data composition (predominantly Hindi in Hinglish lectures).

4.4 Code-Switching Boundary Detection

4.4.1 Frame-Level to Boundary-Level Mapping

Frame-level language labels are converted to language-switch times by detecting transitions:

$$\text{switches} = \{t_j : l[j] \neq l[j-1]\} \tag{30}$$

where $l[j] \in \{0, 1\}$ is the predicted language label for frame j. Each switch frame j corresponds to a timestamp:

$$t_j = j \times \text{hop_seconds} \tag{31}$$

where $\text{hop_seconds} = \frac{\text{hop_length}}{f_s} = \frac{160}{16000} = 0.01$ s (10 ms per frame).

4.4.2 Boundary Matching with Tolerance

Given reference switches $\mathcal{R}$ and predicted switches $\mathcal{P}$, we match switches within a tolerance window τ:

$$\text{matched}(r, p) = |r - p| \leq \tau \tag{32}$$

In our evaluation, $\tau = 0.2$ s (200 ms), a standard tolerance in speech processing literature.

$$\text{TP} = |\{(r, p) : r \in \mathcal{R}, p \in \mathcal{P}, \text{matched}(r, p)\}| \tag{33}$$

4.4.3 Switch-Level Metrics

$$\text{Switch Precision} = \frac{\text{TP}}{|\mathcal{P}|}, \quad \text{Switch Recall} = \frac{\text{TP}}{|\mathcal{R}|} \tag{34}$$

$$\text{Mean Absolute Error (MAE)} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \min_{r \in \mathcal{R}} |r - p| \tag{35}$$

5 Ablation Study: Prosody Warping Impact

5.1 Experimental Design

We compare two TTS synthesis configurations to quantify the impact of prosody transfer:

1. **Baseline (Flat)**: TTS without any prosody transfer. Synthesized speech preserves only Parler TTS's native prosody.

2. **With DTW Warping**: Full prosody transfer using the DTW formulation (Section 3.3.3), aligning source lecture prosody to each TTS chunk.

5.1.1 Data Setup

- **Reference Audio**: 60-second extracted English lecture segment (female native speaker)

- **TTS Chunks**: 20 sentences synthesized via Parler TTS

- **Duration Range**: 2–8 seconds per chunk

- **Evaluation Metric**: Mean Cepstral Distortion (MCD) between reference mel-spectrogram and synthesized mel-spectrogram

- **Mel-scale Parameters**: 128 mel-bands, normalized in dB scale

5.2 Mean Cepstral Distortion (MCD)

MCD measures spectral similarity between reference and synthesized speech:

$$\text{MCD} = \frac{10}{\sqrt{2} \ln 10} \sqrt{\sum_{t=1}^{T} \sum_{c=1}^{C} (c_{\text{ref}}(t) - c_{\text{synth}}(t))^2} \tag{36}$$

where:

- $c_{\text{ref}}(t)$ = cepstral coefficient c at frame t from reference audio

- $c_{\text{synth}}(t)$ = cepstral coefficient c at frame t from synthesized audio

- C = number of cepstral coefficients (typically 13)

- T = number of frames

5.3 Results

Configuration	Mean MCD (dB)	Std Dev MCD (dB)	Improvement
Flat Synthesis (Baseline)	8.24	0.87	—
DTW Prosody Transfer	6.14	0.52	+2.10 dB (**25.5%**)
Linear Fallback (>10min)	6.51	0.68	+1.73 dB (**21.0%**)

Table 2: Ablation Study Results: Prosody Warping Impact on MCD

Key Findings:

1. DTW warping reduces MCD by 2.10 dB (25.5% relative improvement), moving from moderate quality into the high-quality range (4–6 dB).

2. Linear fallback gracefully degrades to 6.51 dB (only 0.37 dB worse than DTW), demonstrating effective memory-safe fallback for long audio.

3. Standard deviation of MCD decreases from 0.87 dB (baseline) to 0.52 dB (DTW), indicating more consistent quality across chunks.

6 End-to-End Pipeline Results

6.1 Pipeline Workflow

The complete speech processing pipeline consists of five stages:

$$\begin{aligned} \text{Raw Audio} &\xrightarrow{\text{Denoise}} \text{Clean Audio} \\ &\xrightarrow{\text{ASR}} \text{Transcript} \\ &\xrightarrow{\text{Translate}} \text{Hinglish Text} \end{aligned} \tag{37}$$

$$\begin{aligned} \text{Hinglish Text} &\xrightarrow{\text{TTS}} \text{Synthesized Chunks} \\ &\xrightarrow{\text{Prosody Transfer}} \text{Final Audio} \end{aligned} \tag{38}$$

6.2 Processing Statistics (10-Minute Lecture)

Pipeline Stage	Output / Metric
Audio Duration	600 seconds
ASR Transcription	1,247 words
Average Words Per Chunk	29.7 words
Translation Chunks	42 segments
TTS Synthesis Time	18 minutes (2.7× real-time)
Prosody Transfer Time	2.3 minutes
Total Pipeline Time	28 minutes

Table 3: Pipeline Stage Outputs and Timing

6.3 LID Frame-Level Performance

Language	Precision	Recall	F1	Frames
English	0.5511	0.8546	0.6736	999
Hindi	0.9942	0.9713	0.9827	23,410
Accuracy		0.9687		24,409
Macro-F1		0.8282		—

Table 4: LID Frame-Level Performance on Evaluation Segment

The model achieves 96.87% frame-level accuracy with excellent Hindi precision (99.42%) and recall (97.13%). English recall is also strong (85.46%), though English precision is lower (55.11%), indicating some false-positive Hindi predictions on English frames.

6.4 Code-Switching Boundary Detection Analysis

Metric	Value	Tolerance	Note
Reference Switch Segments	20	—	Ground truth
Detected Switches (200ms)	0	±200 ms	Frame-level precision insufficient
Switch Precision	0.00	—	No matches within tolerance
Switch Recall	0.00	—	Frame-level detection failed
MAE (where matched)	N/A	—	No successful matches

Table 5: Code-Switching Boundary Detection Results

Analysis: The frame-level classification accuracy (96.87%) does not directly translate to high boundary detection precision. Several factors contribute:

1. **Temporal Quantization**: Frame-level labels are quantized to 10 ms steps; ground-truth boundaries may fall between frames.

2. **Class Imbalance**: With ∼96% of frames being Hindi, occasional English misclassifications scatter predicted boundaries temporally.

3. **Label Smoothing**: Frames near switches may have ambiguous language identity due to speaker prosody or transitions.

Relaxing the matching tolerance to 500 ms (realistic for downstream processing) or implementing explicit code-switching boundary detection (not frame-level classification) would improve results.

7 Implementation Details

7.1 Software Stack

Component	Library/Framework	Version
Deep Learning	PyTorch	2.1.0
Audio Processing	librosa	0.10.0
Signal Processing	scipy	1.11.0
ASR	Whisper (OpenAI)	small
Translation	IndicTrans2	1.1
TTS Backends	Parler TTS, IndicF5	Latest
Language Modeling	NLTK	3.8
Data Processing	pandas, numpy	Latest

Table 6: Software Dependencies

7.2 Pipeline Hyperparameters

Hyperparameter	Value
N-gram LM	
N-gram Order	3 (trigram)
LM Weight (α)	0.25
Length Penalty (β)	0.0
Beam Size	5
Prosody Transfer	
DTW Max Cells	20,000,000
Hop Length (samples)	320 (20 ms)
F0 Range	50–500 Hz
LID Model	
LFCC Bins	80
Frame Hop Length (samples)	160 (10 ms)
LSTM Hidden Size	128
Dropout Rate	0.2
Training Epochs	50
Batch Size	32
Learning Rate	1×10^{-3} (Adam)

Table 7: Pipeline Hyperparameters

7.3 Memory Efficiency

7.3.1 DTW Memory Complexity

Full DTW requires $O(T_x \times T_y)$ memory. For a 10-minute audio at 10 ms framerate:

$$\text{Frames} = \frac{600 \text{ s}}{0.01 \text{ s/frame}} = 60,000 \text{ frames} \tag{39}$$

$$\text{Memory (Full DTW)} = 60,000 \times 60,000 \times 8 \text{ bytes} = 28.8 \text{ GB} \tag{40}$$

Our 20M cell limit:

$$T_{\max} = \sqrt{20,000,000} \approx 4,472 \text{ frames} \approx 44.7 \text{ seconds} \tag{41}$$

For audio longer than ∼45 seconds per chunk, linear fallback is triggered automatically.

7.3.2 LID Inference Memory

$$\begin{aligned} &\text{Memory (CNN+BiLSTM)} = \\ &B \times T \times (128 + 256) \text{ features} \times 4 \text{ bytes} \end{aligned} \tag{42}$$

For batch size $B = 1$ and $T = 6,000$ frames:

$$\text{Memory} = 1 \times 6,000 \times 384 \times 4 = 9.2 \text{ MB} \tag{43}$$

Highly efficient for real-time processing.

8 Adversarial Robustness and Spoof Detection

8.1 Anti-Spoofing Classifier

To ensure the pipeline distinguishes between live human speech and synthetic audio, we implement a countermeasure (CM) system based on Linear Frequency Cepstral Coefficients (LFCC) and a CNN classifier.

8.1.1 LFCC Feature Extraction

LFCC is designed to capture anti-spoofing cues by emphasizing lower frequency components where synthesis artifacts often appear. We extract LFCC using a linear-scale mel-filterbank followed by DCT (Discrete Cosine Transform).

Filterbank Design: We create a linear-scale filterbank with K triangular filters spanning from $f_{\min}$ to $f_{\max}$:

$$\text{Filter}_k(f) = \begin{cases} 0 & \text{if } f < e_k \\ \frac{f - e_k}{e_{k+1} - e_k} & \text{if } e_k \le f \le e_{k+1} \\ \frac{e_{k+2} - f}{e_{k+2} - e_{k+1}} & \text{if } e_{k+1} \le f \le e_{k+2} \\ 0 & \text{if } f > e_{k+2} \end{cases} \tag{44}$$

where $e_0, e_1, \ldots, e_{K+1}$ are uniformly spaced frequency edges in linear scale.

Power Spectrum: The STFT magnitude is converted to power:

$$P(f,t) = |X(f,t)|^2 \tag{45}$$

Filterbank Output: Each frame is filtered:

$$S_k(t) = \sum_f \text{Filter}_k(f) \cdot P(f,t) \tag{46}$$

Log Compression: To compress dynamic range:

$$\log S_k(t) = \log\left(\max(S_k(t), 10^{-10})\right) \tag{47}$$

DCT: Finally, we apply the Type-II Discrete Cosine Transform:

$$\text{LFCC}_c(t) = \sum_{k=0}^{K-1} \log S_k(t) \cos\left(\frac{\pi c(k + 0.5)}{K}\right) \quad \forall c \in [0, C-1] \tag{48}$$

where C is the number of cepstral coefficients retained (typically $C = 20$).

8.1.2 CNN Architecture for Spoof Detection

Our anti-spoofing classifier uses a CNN to detect spoofing artifacts in LFCC spectrograms:

Algorithm 3 Anti-Spoof CNN Architecture

Input: LFCC features $\in \mathbb{R}^{B\times 1\times C\times T}$ (batch, channels, coeffs, frames)
Conv2d(1, 32, kernel_size=3, padding=1)
BatchNorm2d(32) + ReLU
MaxPool2d((2, 2))
Conv2d(32, 64, kernel_size=3, padding=1)
BatchNorm2d(64) + ReLU
MaxPool2d((2, 2))
Conv2d(64, 128, kernel_size=3, padding=1)
BatchNorm2d(128) + ReLU
AdaptiveAvgPool2d((4, 4)) $\rightarrow \mathbb{R}^{B\times 128\times 4\times 4}$
Flatten $\rightarrow \mathbb{R}^{B\times 2048}$
Dropout(0.3)
Linear(2048, 64) + ReLU
Dropout(0.2)
Linear(64, 2) $\rightarrow$ logits for {Bona Fide, Spoof}
Output: Logits $\in \mathbb{R}^{B\times 2}$

8.1.3 Equal Error Rate (EER) and ROC Analysis

The classifier outputs a confidence score $s \in [0, 1]$ for each utterance. For binary classification (Bona Fide vs Spoof), we vary threshold τ and compute:

False Positive Rate (FPR):

$$\text{FPR}(\tau) = \frac{FP(\tau)}{N_{\text{BF}}} = \frac{|\{i : s_i \geq \tau, y_i = 0\}|}{N_{\text{BF}}} \tag{49}$$

where N_{BF} is the number of bona fide samples.

False Negative Rate (FNR):

$$\text{FNR}(\tau) = \frac{FN(\tau)}{N_{\text{Spoof}}} = \frac{|\{i : s_i < \tau, y_i = 1\}|}{N_{\text{Spoof}}} \tag{50}$$

where N_{Spoof} is the number of spoof samples.

Equal Error Rate (EER): The threshold where FPR = FNR:

$$\text{EER} = \min_{\tau} |\text{FPR}(\tau) - \text{FNR}(\tau)| \tag{51}$$

EER is the standard metric for spoof detection; lower values indicate better separation. An EER $< 10\%$ demonstrates robust discrimination.

Area Under the ROC Curve (AUC-ROC):

$$\text{AUC-ROC} = \int_0^1 \text{TPR}(\tau)\, d\text{FPR}(\tau) \tag{52}$$

where $\text{TPR}(\tau) = 1 - \text{FNR}(\tau)$. AUC-ROC ranges from 0 to 1, with 1.0 indicating perfect separation.

8.1.4 Spoof Detection Metrics

True / Predicted	Bona Fide	Spoof	Total
Bona Fide	TN	FP	N_{BF}
Spoof	FN	TP	N_{Spoof}

Table 8: Spoof Detection Confusion Matrix

Additional metrics derived from confusion matrix:

$$\text{APCER} = \frac{FN}{N_{\text{Spoof}}} \tag{53}$$

(Attack Presentation Classification Error Rate)

$$\text{BPCER} = \frac{FP}{N_{\text{BF}}} \tag{54}$$

(Bona fide Presentation Classification Error Rate)

8.2 Adversarial Robustness: FGSM-based Attack

We evaluate the robustness of the LID model against adversarial perturbations using a Fast Gradient Sign Method (FGSM)-inspired attack with noise injection.

8.2.1 Perturbation Formulation

Given an audio signal $y \in \mathbb{R}^N$ with predicted label l_{orig}, we apply random Gaussian-like perturbation scaled by epsilon ϵ:

$$y_{\text{perturbed}} = \text{clip}(y + \epsilon \cdot \eta, -1, 1) \tag{55}$$

where:

- $\eta \in \{-1, +1\}^N$ is random sign noise (deterministic via seed)
- $\epsilon \geq 0$ is the perturbation magnitude
- $\text{clip}(\cdot, -1, 1)$ ensures valid audio range

This is a simplified variant of FGSM suitable for speech signals, where we cannot easily compute gradients through the discrete LID model.

8.2.2 Minimum Epsilon Search

To find the minimum perturbation needed to flip the LID prediction from Hindi to English (or vice versa), we use binary search:

Algorithm 4 Binary Search for Minimum Epsilon

Input: audio y, LID inferencer f, max epsilon $\epsilon_{\max}$, min epsilon $\epsilon_{\min}$
$l_{\text{orig}} \leftarrow f(y)$ # Original prediction
$\epsilon_{\text{upper}} \leftarrow \epsilon_{\min}$
while $\epsilon_{\text{upper}} \leq \epsilon_{\max}$ **do**
  $l' \leftarrow f(\text{clip}(y + \epsilon_{\text{upper}} \cdot \eta, -1, 1))$
  if $l' \neq l_{\text{orig}}$ **then**
    break
  end if
  $\epsilon_{\text{upper}} \leftarrow 2 \times \epsilon_{\text{upper}}$
end while
if $\epsilon_{\text{upper}} > \epsilon_{\max}$ and $l' = l_{\text{orig}}$ **then**
  return None (no flip found)
end if
$\epsilon_{\text{lower}} \leftarrow \epsilon_{\text{upper}}/2$
for $k = 1$ to search_steps **do**
  $\epsilon_{\text{mid}} \leftarrow 0.5 \times (\epsilon_{\text{lower}} + \epsilon_{\text{upper}})$
  $l_{\text{mid}} \leftarrow f(\text{clip}(y + \epsilon_{\text{mid}} \cdot \eta, -1, 1))$
  if $l_{\text{mid}} = l_{\text{orig}}$ **then**
    $\epsilon_{\text{lower}} \leftarrow \epsilon_{\text{mid}}$
  else
    $\epsilon_{\text{upper}} \leftarrow \epsilon_{\text{mid}}$
  end if
end for
return ϵ_{upper}

8.2.3 Signal-to-Noise Ratio (SNR) Constraint

To ensure perturbations remain inaudible to humans, we measure SNR in dB:

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \left(\frac{\sum_n y[n]^2}{\sum_n (y_{\text{perturbed}}[n] - y[n])^2} \right) \quad (56)$$

where:

- Numerator: signal power (original audio)
- Denominator: noise power (perturbation)

8.3 Experimental Results

8.3.1 Anti-Spoofing Evaluation

We trained the anti-spoof CNN on 8-second chunks of bona fide (live) and spoof (TTS) audio with 1-second overlaps. Training data: 200 bona fide clips, 200 spoof (Parler TTS) clips. Test set: 64 bona fide, 64 spoof utterances.

True / Pred.	Bona Fide	Spoof	Recall	Total
Bona Fide	64	0	1.0000	64
Spoof	0	64	1.0000	64
Precision	1.0000	1.0000	—	—

Table 9: Anti-Spoof Confusion Matrix (Test Set)

Metric	Value
Equal Error Rate (EER)	0.00%
EER Threshold	0.50
AUC-ROC	1.0000
Accuracy @ 0.5	1.0000
Precision @ 0.5	1.0000
Recall @ 0.5	1.0000
F1-Score @ 0.5	1.0000
APCER	0.00% (FN rate)
BPCER	0.00% (FP rate)

Table 10: Anti-Spoof Metrics

Findings: The anti-spoof classifier achieves perfect separation (EER = 0.0%, AUC = 1.0) between bona fide and TTS-synthesized audio. This demonstrates that Parler TTS output exhibits clear spectral differences from natural speech, detectable via LFCC features. The model generalizes well to the test set, showing no overfitting.

8.3.2 Adversarial Robustness Analysis

We evaluated the LID model's robustness to adversarial noise using the binary search attack on 5-second audio clips. Test set: 20 Hindi clips, 20 English clips. For each clip, we computed the minimum epsilon required to flip the language prediction.

Metric	Hindi→Eng	Eng→Hindi	Mean	Unit
Min ϵ (Mean)	3.84×10^{-4}	4.12×10^{-4}	3.98×10^{-4}	—
Min ϵ (Median)	3.52×10^{-4}	3.78×10^{-4}	3.65×10^{-4}	—
Min ϵ (Std Dev)	1.23×10^{-4}	1.45×10^{-4}	1.34×10^{-4}	—
SNR @ Min ϵ (Mean)	52.3 dB	50.8 dB	51.5 dB	dB
SNR @ Min ϵ (Min)	48.1 dB	47.3 dB	47.3 dB	dB
SNR @ Min ϵ (Max)	56.2 dB	54.9 dB	56.2 dB	dB

Table 11: Adversarial Robustness: Minimum Epsilon to Flip LID

Key Observations:

1. **Low ϵ Threshold**: Mean minimum epsilon is 3.98×10^{-4} (very small). This indicates the LID model is vulnerable to small, carefully crafted perturbations.

2. **High SNR at Flip**: Despite low epsilon, SNR remains > 47.3 dB (well above the 40 dB imperceptibility threshold). This means perturbations are inaudible to humans while still causing misclassification.

3. **Language Asymmetry**: Hindi→English flips require slightly lower epsilon (3.84×10^{-4}) than English→Hindi (4.12×10^{-4}), suggesting the CNN+BiLSTM model has a slight bias toward English misclassification or higher sensitivity to Hindi-specific acoustic cues.

4. **Consistency**: The relatively low standard deviation ($\sim 1.34 \times 10^{-4}$) suggests consistent vulnerability across different input utterances.

9 Discussion

9.1 Strengths

1. **Mathematically Grounded**: Each component has rigorous mathematical formulation (Eqs. 4–20) enabling reproducibility and future optimization.

2. **Memory Efficiency**: DTW guard with linear fallback enables long-form audio processing on edge devices without GPU memory exhaustion.

3. **Modular Architecture**: Components are independently trained (ASR, LID, TTS) and can be swapped without retraining the full pipeline.

4. **Prosody Transfer Gains**: Ablation study quantifies 2.1 dB MCD improvement, demonstrating clear naturalness benefit.

5. **Robust Encoding**: Laplace smoothing ensures non-zero probabilities, graceful degradation on OOV phrases in constrained language models.

9.2 Limitations

1. **LID Class Imbalance**: Hindi precision (99.42%) vastly exceeds English precision (55.11%). Training on balanced Hindi-English corpora would mitigate this.

2. **Code-Switching Timing**: Frame-level classification (10 ms resolution) insufficient for exact boundary detection. Explicit boundary detection models needed.

3. **Linear Fallback Quality**: Linear interpolation (Eq. 17) is crude approximation; more sophisticated fallback (e.g., piecewise linear, spline) possible but not implemented.

4. **Single-Speaker ASR LM**: N-gram LM trained on provided corpus; diverse speaker variability not modeled.

10 Conclusion

This work presents a complete end-to-end Hinglish speech processing pipeline with rigorous mathematical formulation and practical implementation. We have demonstrated:

The ablation study validates the prosody contribution, and implementation details enable reproducibility. Future work will focus on improving code-switching temporal precision, cross-domain generalization, and streaming inference.

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